



🎓 COMPLETE CURRICULUM · 6 MONTHS

Python + Machine Learning

Beginner to Advanced — Full Course Syllabus

24

MODULES

50+

PROJECTS

500+

INTERVIEW Q'S

22

TOOLS



Python
Fundamentals



Data
Science



Statistics
& Prep



Machine
Learning



Advanced
AI



Deploy &
Capstone

Prerequisites: Basic computer knowledge — no programming experience required

Duration: 6 Months · **Format:** Hands-on with 50+ projects

Target Roles: Python Developer · Data Analyst · Data Scientist · ML Engineer

● PHASE 1 — PYTHON PROGRAMMING

Weeks 1–10

- 01
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Introduction to Programming

What is Programming · How Computers Execute Code · Installing Python · IDE Setup (VS Code, PyCharm, Jupyter Notebook) · Python Interpreter · Writing Your First Program · Comments · Variables · Data Types · Input and Output · Type Conversion
- 02
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Python Fundamentals

Arithmetic, Assignment, Comparison, Logical, Identity, Membership, Bitwise Operators · Conditional Statements (if, else, elif, Nested if) · Loops (for, while, break, continue, pass)
- 03
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Python Data Structures

Strings (Indexing, Slicing, Methods, Formatting, f-Strings) · Lists (Creation, Indexing, Nested, Methods, Comprehensions) · Tuples (Packing, Unpacking) · Sets (Union, Intersection, Difference) · Dictionaries (Keys, Values, Items, Nested)
- 04
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Functions & Modules

Parameters & Arguments · Return Statement · Default Parameters · Keyword Arguments · Lambda Functions · Recursion · Built-in Modules · Custom Modules · Packages · pip · Virtual Environments
- 05
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Object-Oriented Programming

Classes · Objects · Constructors · Instance Variables · Methods · Encapsulation · Inheritance · Polymorphism · Abstraction
- 06
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Exception Handling & File Handling

try, except, else, finally, raise · Custom Exceptions · Read Files · Write Files · Append Files · CSV Files · JSON Files
- 07
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Advanced Python

Decorators · Iterators · Generators · Context Managers · Regular Expressions · Collections Module · Datetime Module · OS Module · Sys Module · Logging

● PHASE 2 — DATA SCIENCE

Weeks 11–16

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SQL for Data Science

Database Basics · SQL Installation · CRUD Operations · Constraints · Aggregate Functions · GROUP BY · HAVING · Joins · Subqueries · Views
- 09
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Git & GitHub

Git Installation · Git Basics · Repository Management · Branching · Merging · Pull Requests · GitHub Workflow
- 10
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NumPy

Arrays · Array Operations · Indexing · Broadcasting · Reshaping · Statistics · Random Module · Linear Algebra
- 11
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Pandas

Series · DataFrames · Reading CSV/Excel Files · Data Cleaning · Missing Values · GroupBy · Merge · Pivot Tables · Filtering
- 12
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Data Visualization

Matplotlib (Line Plot, Scatter Plot, Histogram, Pie Chart, Bar Chart) · Plotly (Interactive Charts, Dashboards, Time Series Visualization)

● PHASE 3 — STATISTICS & ML PREPARATION

Weeks 17–19

- 13
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Statistics for Machine Learning

Mean, Median, Mode, Variance, Standard Deviation · Probability · Conditional Probability · Bayes Theorem · Probability Distributions · Hypothesis Testing · Z-Test · T-Test · Chi-Square Test

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Data Preprocessing
- Handling Missing Values · Removing Duplicates · Outlier Detection · Feature Scaling (Normalization, Standardization) · Label Encoding · One-Hot Encoding · Feature Engineering · Feature Selection
- 15
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Machine Learning Fundamentals
- AI vs Machine Learning vs Deep Learning · Types of Machine Learning · Training vs Testing · Overfitting · Underfitting · Bias-Variance Tradeoff · Cross Validation

● PHASE 4 — MACHINE LEARNING

Weeks 20–23

- 16
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Supervised Learning
- Regression:** Linear Regression · Multiple Linear Regression · Polynomial Regression · **Classification:** Logistic Regression · K-Nearest Neighbors (KNN) · Naive Bayes · Decision Trees · Random Forest · Support Vector Machine (SVM)
- 17
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Ensemble Learning
- Bagging · Boosting · AdaBoost · Gradient Boosting · XGBoost · LightGBM · CatBoost
- 18
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Unsupervised Learning
- Clustering:** K-Means · Hierarchical Clustering · DBSCAN · **Dimensionality Reduction:** PCA · t-SNE
- 19
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Model Evaluation
- Regression Metrics:** MAE · MSE · RMSE · R^2 Score · **Classification Metrics:** Accuracy · Precision · Recall · F1 Score · ROC Curve · Confusion Matrix

● PHASE 5 — ADVANCED AI

Week 24+

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Natural Language Processing
- Text Preprocessing · Tokenization · Lemmatization · Stemming · Stopword Removal · TF-IDF · Word Embeddings · NLTK · spaCy
- 21
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Time Series Analysis
- Time Series Concepts · Moving Average · Seasonal Decomposition · ARIMA · Forecasting Techniques
- 22
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Deep Learning
- Artificial Neural Networks (ANN) · Activation Functions · Forward Propagation · Backpropagation · TensorFlow · Keras · Convolutional Neural Networks (CNN) · Recurrent Neural Networks (RNN) · LSTM

● PHASE 6 — DEPLOYMENT & CAPSTONE

Week 24+

- 23
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Model Deployment
- Model Serialization (Pickle, Joblib) · Flask · FastAPI · REST APIs · Docker Basics · Cloud Deployment Basics (AWS / GCP / Azure)
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Capstone Projects
- House Price Prediction · Movie Recommendation System · Credit Card Fraud Detection · Customer Churn Prediction · Loan Approval Prediction · Email Spam Detection · Face Mask Detection · Stock Price Prediction · Resume Screening System · Sentiment Analysis Dashboard · Sales Forecasting System · Medical Disease Prediction · Fake News Detection · Image Classification Web App

Tools & Libraries Covered

Python	Jupyter	VS Code	Git & GitHub	SQL	NumPy
Pandas	Matplotlib	Plotly	scikit-learn	NLTK	spaCy
TensorFlow	Keras	Flask	FastAPI	Docker	OpenCV
XGBoost	LightGBM	CatBoost	PyCharm		

Interview Preparation

300+

Python Coding Questions

100+

SQL Interview Questions

100+

ML Interview Questions

ML System Design

Building scalable ML pipelines, data architecture patterns, and production deployment strategies for real-world applications.

Resume & Portfolio

Resume building workshops, GitHub portfolio creation, and strategies for showcasing capstone projects to stand out to recruiters.

Mock Technical Interviews & Aptitude

Simulated technical interviews with detailed feedback, plus aptitude and logical reasoning preparation to build confidence for real interview settings.

Learning Outcomes

- ✓ Write clean, efficient, and production-ready Python programs using best practices and modern syntax.
- ✓ Use SQL for database operations and Git/GitHub for version control in real-world team workflows.
- ✓ Analyze, clean, transform, and visualize complex datasets using Pandas, NumPy, Matplotlib, and Plotly.
- ✓ Build, evaluate, and tune machine learning models — from linear regression to ensemble methods like XGBoost.
- ✓ Develop deep learning applications using TensorFlow and Keras, including CNNs and RNNs for image and sequence data.
- ✓ Build NLP pipelines for text classification, sentiment analysis, and information extraction using NLTK and spaCy.
- ✓ Deploy machine learning models as REST APIs using Flask and FastAPI, with Docker and cloud deployment basics.
- ✓ Present a portfolio of 14+ end-to-end projects for Python Developer, Data Analyst, Data Scientist, and ML Engineer roles.

TARGET ROLES AFTER COMPLETION

- Python Developer
- Data Analyst
- Data Scientist
- ML Engineer